MAGNETOSTATICS

SUB-TOPICS

Retentivity : Oa

Coercivity : Ob

- Magnetic Dipole
- Magnetic Susceptibility
- Curse Law.
- What is Rheostat.
- Geographical Meridian, Magnetic Meridian } : Earths Magnetic field.
  i. Magnetic Declination (θ)
  ii. Angle of Dip (δ)
  iii. Apparent Dip (ϕ)
- Bar Magnets.
- Time Period of Bar Magnet
- Vibration Magnetometer.
- Hysterius Law :

Concepts & Key Points...

1. Magnetic Dipole $|\bar{M}| = NiA$

2. $\phi = \int \bar{B} \cdot d\bar{s}$

3. Magnetic Flux Density : $= \frac{\phi}{A}$

$$\phi = \bar{B}\bar{S}$$

4. $B = \mu_0 n i$

5. X_m (Magnetic Susceptibility) $= \frac{I}{H}$ $H = ni$

6. $B_0 = \mu_0 H$ & $B_m = \mu_0 i$

$$B = B_0 + B_m = \mu_0 (H + I) \quad * * *$$

7. $\mu_r = 1 + X_m$

Magnetic Permeability, Magnetic Susceptibility

$$\bar{\tau} = \bar{M} \times \bar{B}$$

8. Work Done : $MB(\cos\theta_1 - \cos\theta_2)$

9. $T = 2\pi\sqrt{\frac{I}{MB}}$

10. $T = 2\pi\sqrt{\frac{I}{MB_H}}$

11. $i = \left[\frac{2RB_H}{\mu_0 N}\right] \tan\theta$ $i = K\tan\theta$ $i \propto \tan\theta$.

$$\tan\delta = \tan\phi \cos\beta$$

12. $|\bar{P}| = q(l)$

13. Angle Of Dip (θ) then $\boxed{B_H = B_V \tan 30^\circ}$ $\tan\delta = \frac{B_V}{B_H}$

Another B Value (Magn) One B Value (Magnetic Moment)

Where

B_V : Earth's Vertical

B_H : Earth's Horizontal Magnetic field lines...

14. Magnetization

$$(M) = \frac{\text{magnetic moment}}{\text{Unit Vol}}$$

$$(M) = \frac{m}{V}$$

15. Magnetic intensity $(H) = \frac{B}{\mu} - M$.

16. Magnetic Susceptiblity $(\chi_m) = \frac{I}{H}$

17. $\boxed{H = ni}$

18. $B = \mu_0 (H + i)$.

19. Magnetic Permiability = 1 + Magnetic Susceptiblity...

Units of I:

$$I = \frac{Amp\text{-}m^2}{m^3} = \frac{Amp}{m}$$

$$I = \frac{\text{Magnetic Moment}}{\text{Volume}} = \frac{M}{V}$$

Note: SI Units of H & I are Same: $H \rightarrow \frac{Amp}{m}$, $I \nearrow$ Units

Magnetic Susceptibility (X_m):

: Ratio of intensity of Magnetization (I) induced in the Material to the Magnetizing force (H) applied on it.!

$\Rightarrow$ $\boxed{X_m = \frac{I}{H}}$

$\Rightarrow$ No Units.

$\boxed{H = ni}$

Air : $B_0 = \mu_0 ni$

Medium (1) : $B_1 = \mu_1 ni$

Medium (2) : $B_2 = \mu_2 ni$

$B_0 = \mu_0 H$

$B_1 = \mu_1 H$

$B_2 = \mu_2 H$.

Relation b/w Magnetic Permeability and Magnetic Susceptibility:

$B_0 = \mu_0 H$ $\quad$ $B_m = \mu_0 I$

$B = B_0 + B_m$

$\boxed{B = \mu_0 (H + I)}$

$X_m = \frac{I}{H}$ $\quad$ $I = X_m H$

$$B = \mu_0 (H + X_m H) = \mu_0 H (1 + X_m)$$

$$B = \mu H$$

$\mu H = \mu_0 H (1 + X_m)$ $\quad$ $\frac{\mu}{\mu_0} = 1 + X_m$

$\boxed{\mu_r = 1 + X_m}$

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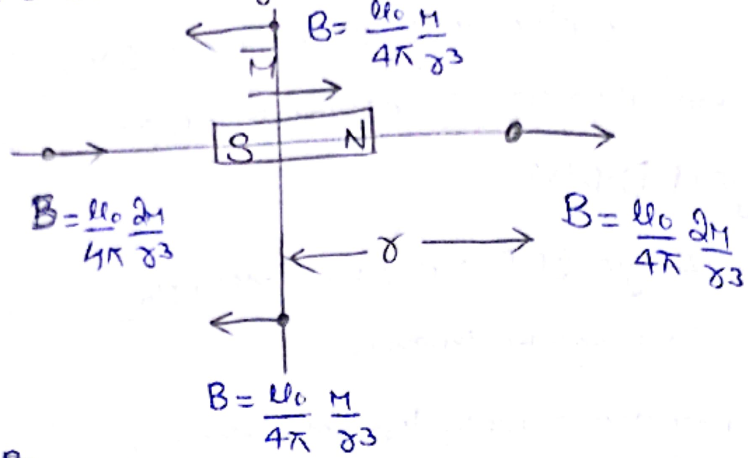
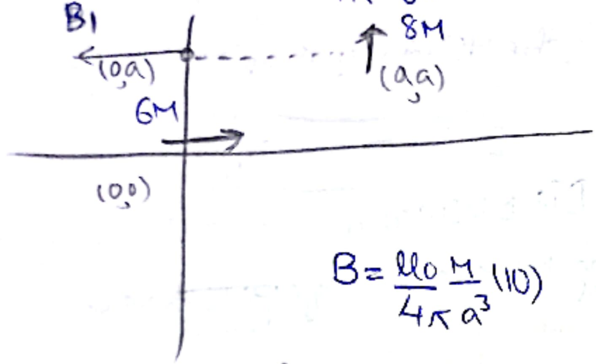
Sub	X_m	μ_r	μ
Dia	$-1 \le X_m < 0$	$0 \le \mu_r < 1$	$\mu < \mu_0$
Para	$0 < X_m < \epsilon_0$	$1 < \mu_r < (1+\epsilon)$	$\mu > \mu_0$
Ferro	$X_m >> 1$	$\mu_r >> 1$	$\mu >> \mu_0$

: Relation b/w Relative Magnetic Permeability, Magnetic Susceptibility..

$\bar{B}$ due to Short bar Magnet :



1.



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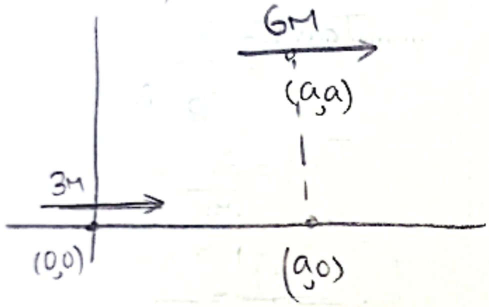
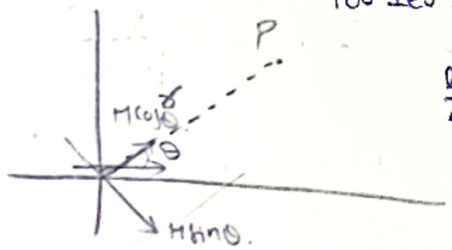


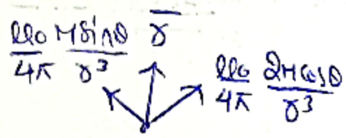
1. Find $\bar{B}$ at (a,0) : Zero

2 Times : For axial , Director remains Same

1 Times : For ⊥er bisector, 180° Turn ...

3.





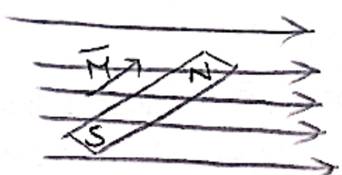
- Since P is not as ⊥er bisector or axial line ....

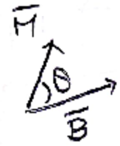
$$\bar{r} = \frac{\mu_0}{4\pi}\frac{M}{r^3}\left(\sqrt{(2\cos\theta)^2+(\sin\theta)^2}\right)$$

$$\bar{r} = \frac{\mu_0 M}{4\pi r^3}\left(\sqrt{1+3\cos^2\theta}\right)$$

Bar Magnetic Placed in a Uniform Magnetic field :

1. Net Force Zero.
2. Net τ : $\bar{M}\times\bar{B}$





Potential Energy : $-\bar{M}\cdot\bar{B}$

Work Done to rotate Magnetic Dipole (Bar Magnetic)

Force θ_1 to θ_2: $\Delta WD = WD_{final} - WD_{initial}$.

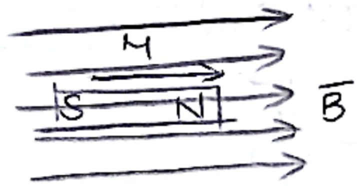
$$W = U_f - U_i$$

$$W = -MB\cos\theta_2 - (-MB\cos\theta_1)$$

$$W = MB(\cos\theta_1 - \cos\theta_2).$$

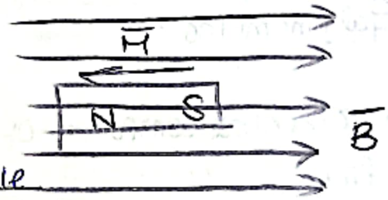
Case I : θ° = Zero

$F = 0$, $T = 0$ } $U = -MB$, Stable eq



Case II : $\theta^\circ = 180^\circ$

$F = 0$, $T = 0$ } $U = MB$, Unstable equilibrium...

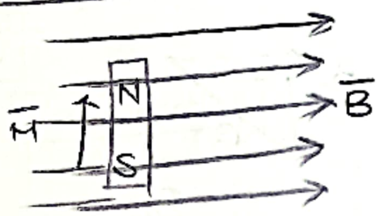


Case III : $\theta^\circ = 90^\circ$

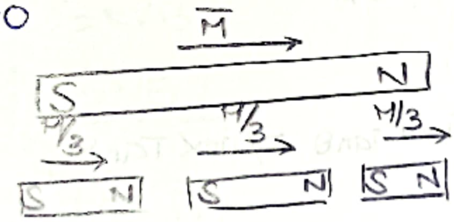
$F = 0$

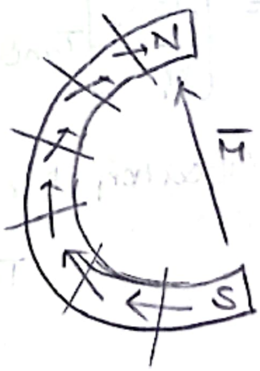
$T = MB$

$U = 0$



Cut :

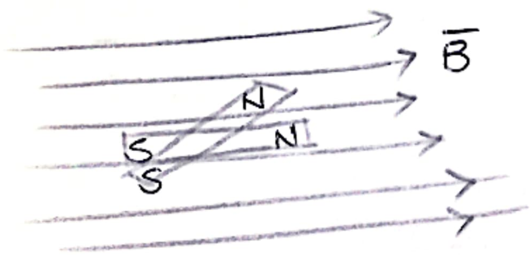




$\bar{M} = (m)(l)$

Time Period of Bar Magnet

Oscillaty in a UMF :



$$T = -MB\sin\theta.$$

As θ is very small. $T = -MB\theta$

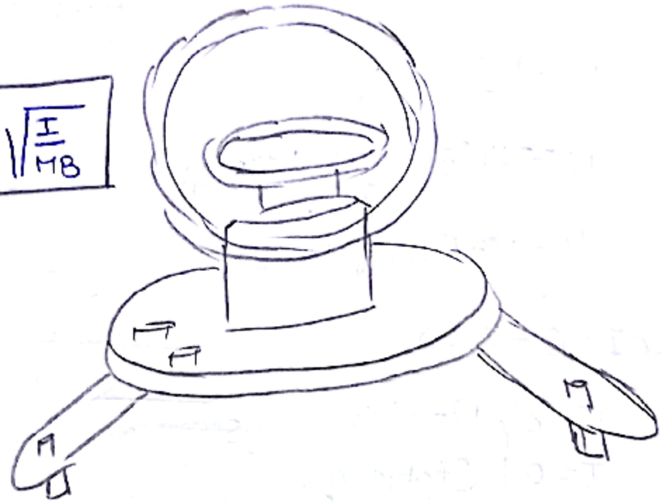
$$I\alpha - MB\theta. \quad \boxed{\alpha = -\left(\frac{MB}{I}\right)\theta}$$

$$a = -\omega^2 x$$

$$\omega = \sqrt{\frac{MB}{I}}$$

$$T = \frac{2\pi}{\omega}$$

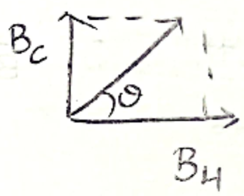
$$\boxed{T = 2\pi\sqrt{\frac{I}{MB}}}$$

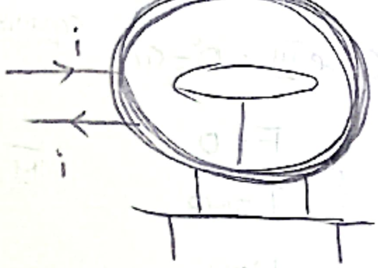


Tangent Galvanometer:

B_H = Horizontal component of Earth Magnetic field.

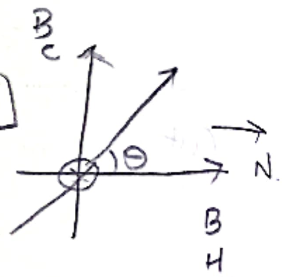
$$\tan\theta = \frac{B_C}{B_H}$$





$$\tan\theta = \frac{\mu_0 N i}{2RB_H}$$

$$i = \left[\frac{2RB_H}{\mu_0 N}\right]\tan\theta \qquad i = K\tan\theta \Rightarrow \boxed{i \propto \tan\theta}$$



Vibration Magnenometer:

$$T = 2\pi\sqrt{\frac{I}{MB_H}}.$$